Collective Robotics
Part 6: Collective decisions

Prof. Dr. Javad Ghofrani

Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

Introduction

Decision making

Decision:

selection between possible actions

Decision making:

cognitive process of selecting an action from a number of alternatives

Optimal decision:

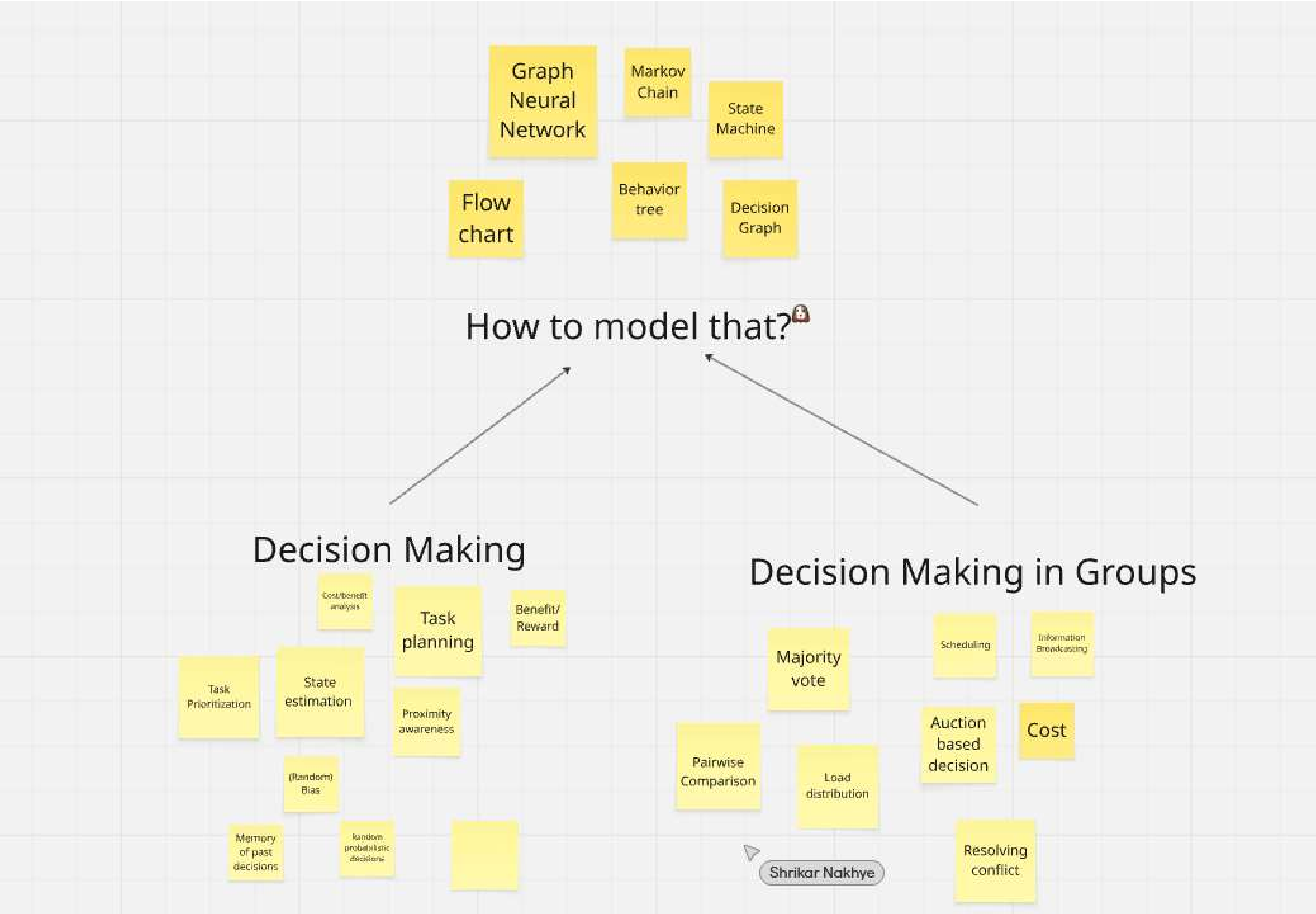
no alternative decision results in a better outcome

Today: How to model decision making?

- How we make decisions?
- Is there any difference between decision making in individual and group levels?
- How can we model the decision making for individuals and groups?
- How can we implement group decision making in robots?

5 Minutes of Thinking

- Use miro-board for sharing your thoughts



Graph Neural Network
Markov Chain
State Machine
Flow chart
Behavior tree
Decision Graph
How to model that?
Decision Making
Cost/benefit analysis
Task planning
Benefit/ Reward
Task Prioritization
State estimation
Proximity awareness
(Random) Bias
Memory of past decisions
Random probabilistic decisions
Decision Making in Groups
Majority vote
Scheduling
Information Broadcasting
Auction based decision
Cost
Pairwise Comparison
Load distribution
Resolving conflict
Shrikar Nakhye

Decision making under uncertainty

the actual art of decision making,

we need to know preferences between different possible outcomes,

we define a utility of outcomes (quality of being useful)

rational decision:

an optimal decision under a set of constraints (e.g., uncertainty),

an agent is rational if and only if it chooses the action that yields the highest expected utility, averaged over all the possible outcomes of the action (principle of maximum expected utility)

(Russell and Norvig, 1995)

Example: Decision Tree

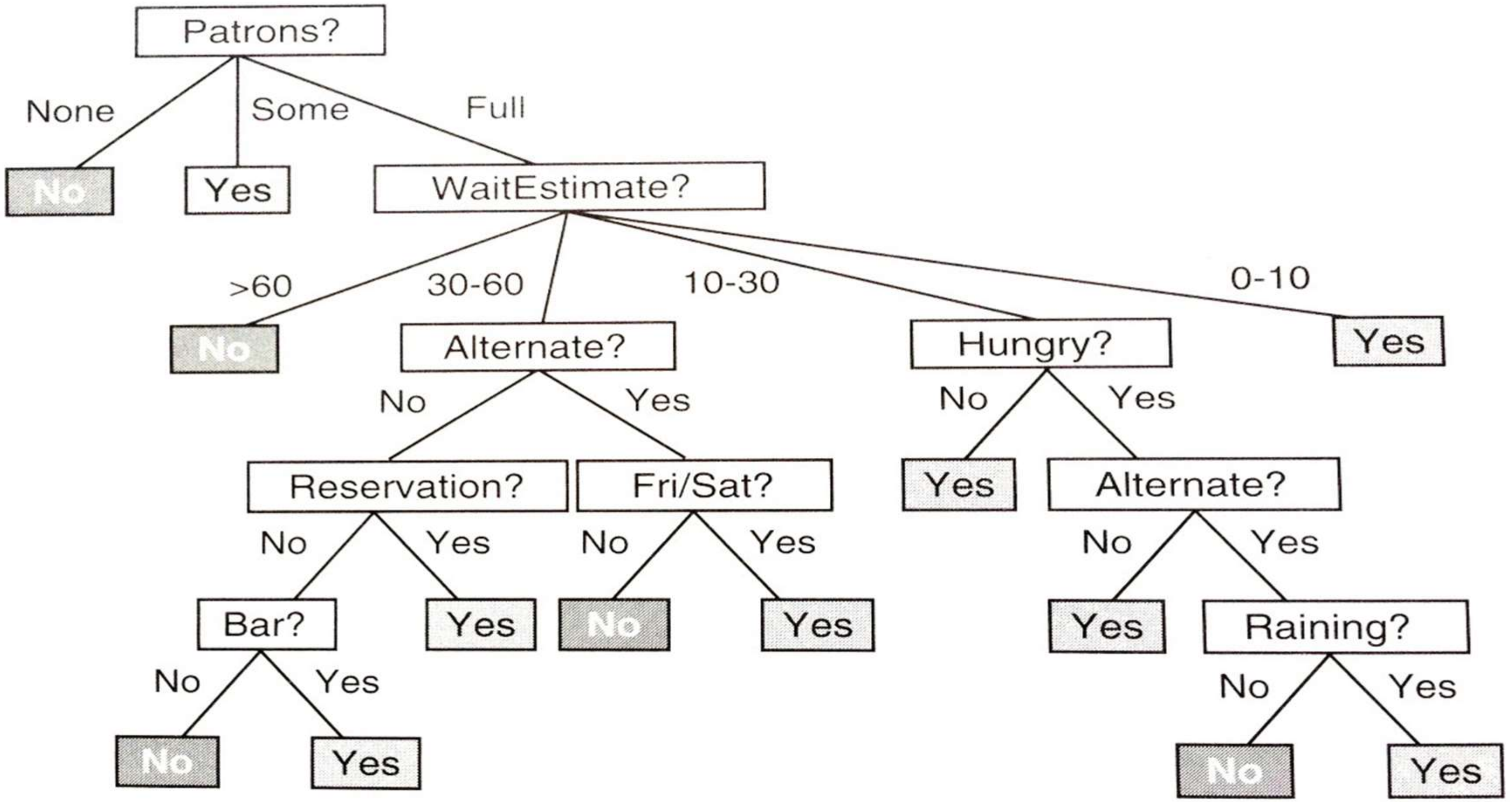


Figure 18.2 A decision tree for deciding whether to wait for a table.

A general model of decision-making

model

decider or player S

m possible alternative actions $A_1, \ldots, A_m$

n possible initial conditions/states $s_1, \ldots, s_n$

utility $N_{ij} \in R$ for action A_i in state s_j

utility matrix $N = (N_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$

strategy under certainty about initial state (i.e., given j) is clear:

select $\hat{\imath} \in \{1, \ldots, m\}$ so that $N_{\hat{\imath}\, j} = max_i\, N_{ij}$

Strategies under uncertainty

uncertainty: we are not sure about the initial condition but might know probabilities of being in a certain state

- cautious strategy (max-min-utility)
- high-risk strategy (max-max-utility)
- alternatively cautious strategy (min-max-utility)

Strategies under uncertainty

$$\begin{array}{cc} & \begin{array}{ccc} S1 & S2 & S3 \end{array} \\ \begin{array}{c} A1 \\ A2 \end{array} & \begin{bmatrix} N_{11} & N_{12} & N_{13} \\ N_{21} & & N_{23} \end{bmatrix} \end{array}$$

Cautious Strategy (max-min-utility)

select $\hat{i} \in \{1, \dots, m\}$ so that $N_{\hat{i}\, j} = max_i min_j\, N_{ij}$

the minimally possible utility (i.e., assumed worst possible state j) is maximized over all possible strategies i

High-risk strategy (max-max-utility)

select $\hat{i} \in \{1, \dots, m\}$ so that $N_{\hat{i}\, j} = max_i max_j\, N_{ij}$

the maximally possible utility (i.e., assumed best possible state j) is maximized over all possible strategies i

Alternatively cautious strategy (min-max-utility)

select $\hat{i} \in \{1, \dots, m\}$ so that $R_{\hat{i}\, j} = min_i max_j\, R_{ij}$

For $N_j = max_i\, N_{ij}\ (maximal\ utility\ in\ state\ j)$

$R_{ij} = N_i - N_{ij}\ (risk\ of\ action\ i\ in\ state\ j)$

the maximally possible risk (i.e., assumed most risky state j) is minimized over all possible strategies i

Strategies under uncertainty: Example

Given $m = n = 2, \quad N_{11} = 0, \quad N_{21} = N_{22} = 1, N_{12} = 100$
utility matrix: N = ?
Risk matrix: R= ?

cautious strategy: ?
high-risk strategy: ?
alternatively cautious strategy: ?

notation: N_{ij}
column-wise: states j
row-wise: actions i

Strategies under uncertainty: Example

Given $m = n = 2, \quad N_{11} = 0, \quad N_{21} = N_{22} = 1, N_{12} = 100$

utility matrix: N = $\begin{pmatrix} 0 & 100 \\ 1 & 1 \end{pmatrix}$

$N_1 = 1, N_2 = 100$

Risk matrix: R= $\begin{pmatrix} 1 & 0 \\ 0 & 99 \end{pmatrix}$

cautious strategy: A_2

high-risk strategy: A_1

alternatively cautious strategy: A_1

notation: N_{ij}

column-wise: states j

row-wise: actions i

Group decision making

Group/collaborative decision making:

a group of individuals makes a choice from alternatives; effects of social influence, group polarization

Consensus decision-making: avoid ‘winners’ and ‘losers’, a majority approves the decision while the minority agrees to go along, i.e. minority could veto

Voting-based methods: majority (> 50%), plurality (largest block decides, even if < 50%)

Group decision making – challenges

Groupthink (1972, Irving Janis):

psychological phenomenon within a group of people who minimize conflict and reach a consensus decision without critical evaluation of alternative ideas

How to get from individual decisions to a decision of the group?

⇒ voting paradox! (next)

Preparation: Relation

set theory/logic, a relation assigns truth values to k-tuples depending on whether a certain property does or does not hold

A relation R over a domain $X : R \subseteq X^2 = X \times X$

we write $(x, y) \in R$ or just xRy

properties of relations: reflexive: xRx for all x

transitive: if xRy and yRz then xRz

symmetric: if xRy then yRx

connex: if not xRy then yRx (general comparability)

Majority decision

each voter is represented by a relation R_i

$N(x, y)$ is number of voters with xRy

majority relation M: $xMy \Leftrightarrow N(x, y) \geqslant N(y, x)$

M is always reflexive and connex (counting votes always works) but not always transitive!

example of a voting paradox

Candidates $\mathrm{X} = \{a, b, c\}$, Voters $I = \{1, 2, 3\}$

$$R1 = \{(a, b), (b, c), (a, c), (a, a), (b, b), (c, c)\}$$
$$R2 = \{(c, a), (a, b), (c, b), (a, a), (b, b), (c, c)\}$$
$$R3 = \{(b, c), (c, a), (b, a), (a, a), (b, b), (c, c)\}$$

The majority relation has a cycle

$$M = \{(a, b), (b, c), (c, a), (a, a), (b, b), (c, c)\}$$

Social choice theory (also known as: theory of collective choice)

Analysis of how to aggregate individual preferences to achieve a group decision

Voting paradox (Marquis de Condorcet, 1785)

Arrow's impossibility theorem (Kenneth Arrow, 1951, Nobel laureate)
With more than 2 candidates no rank order voting system can convert the ranked preferences of individuals into a community-wide ranking while also meeting a specific set of (democratic) criteria.

Ostrogorski paradox (Hans Daudt and Douglas W. Rae, 1976)
distortion of voters' attitudes when voting for collections of factual issues in the form party platforms

Group decision-making in animals

Group decisions are complex and often paradoxical animals often need a consensus decision

Examples: nest site/resting place selection, heading of a flock

How can animals possibly achieve consensus decisions efficiently?

simplifying factors:

- typically no conflicting interests
- (“what’s good for the swarm is good for all”)
- approximate decisions are often sufficient
- a slow iterative process might be acceptable

Consensus decision making in animals

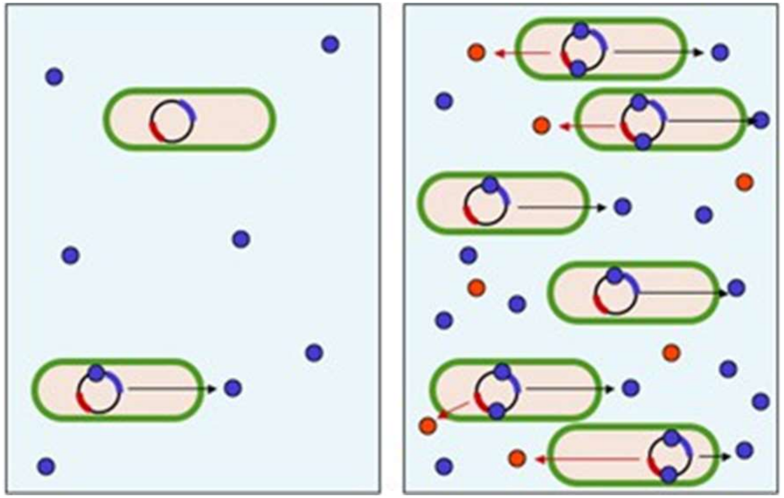
Nest choice in bees,

only 5% disperse as scouts and explore potential nest sites, they return, and advertise their findings by dancing to other scouts, eventually the scouts reach a consensus; the remaining 95% of the swarm do not contribute

Travel routes in navigating birds,

limited knowledge, more experienced birds might contribute more but also majority decisions might occur, self-organizing decision processes seem likely in large groups

Quorum sensing in bacteria



Quorum:

minimum number of members of a deliberative assembly necessary to conduct the business of that group

Quorum sensing:

ability of unicellular organisms to measure the cell density of the population; used for coordination of processes that are only efficient if many do it at the same time

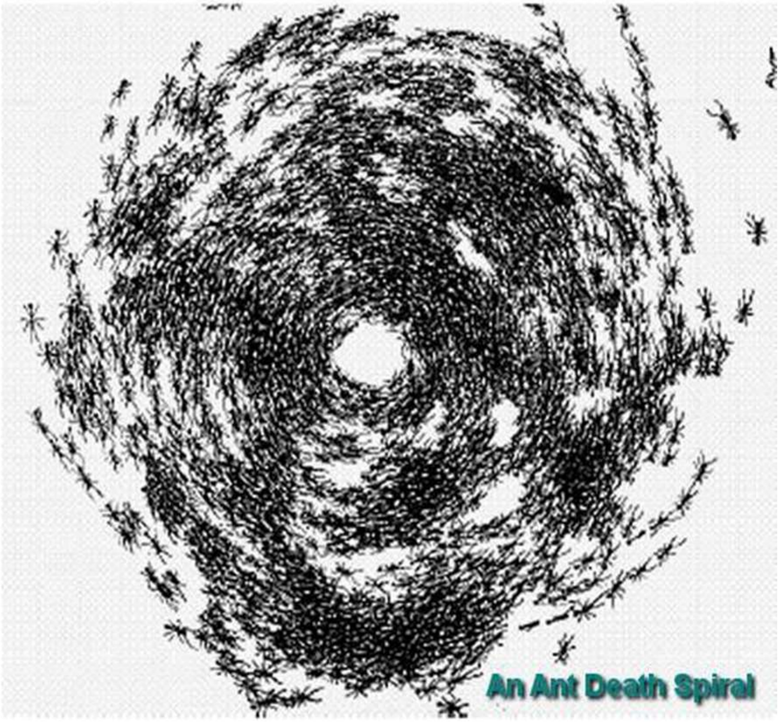
Quorum sensing in decentralized systems

mechanism:

a bacterium always secretes a signaling molecule (SM), how much is produced depends on how much of the SM is detected; once the concentration of SM passes a threshold, more SM is synthesized (pos. feedback loop)

Quorum sensing can function as decision-making process in any decentralized system

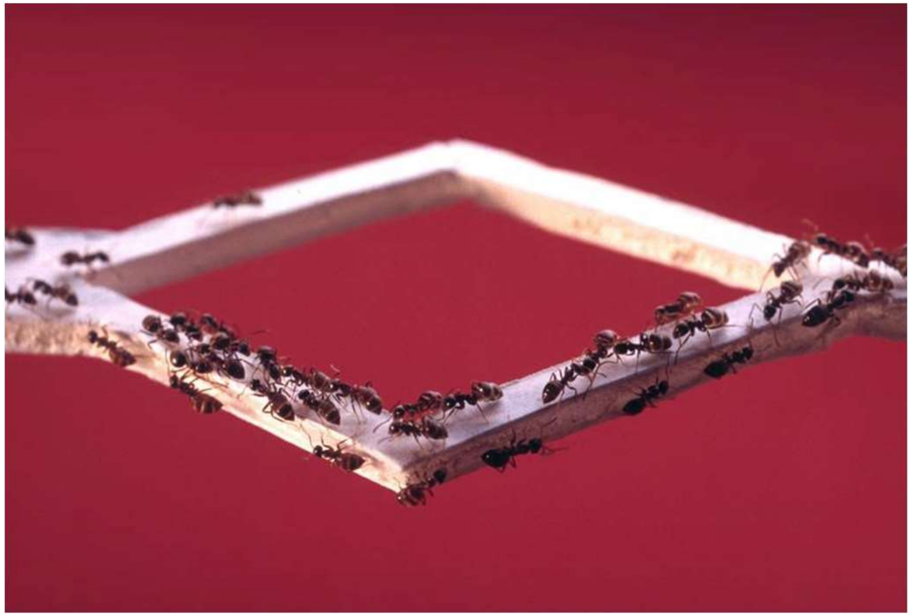
Example for bad collective decision



https://www.youtube.com/watch?v=BPEfeHRP_8M

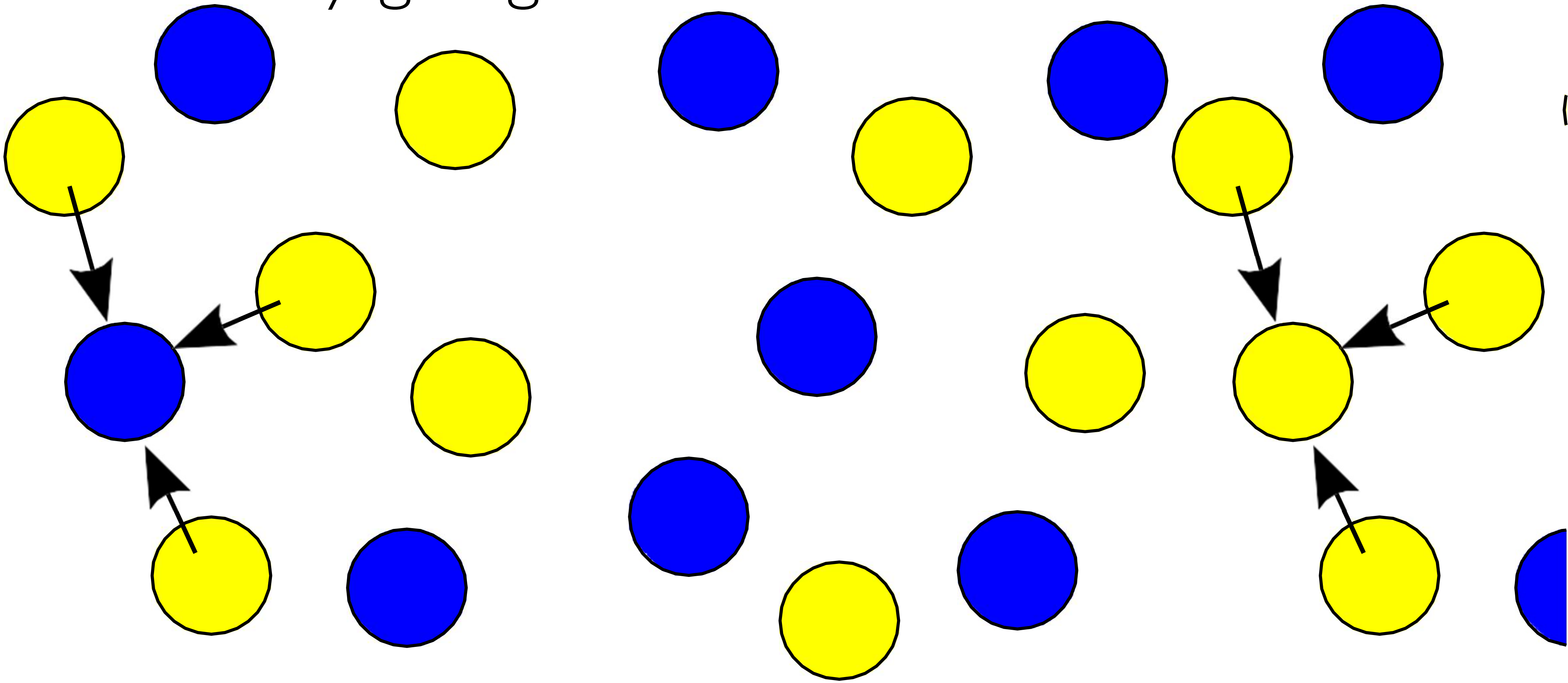
circular milling

Collective decision making in swarms

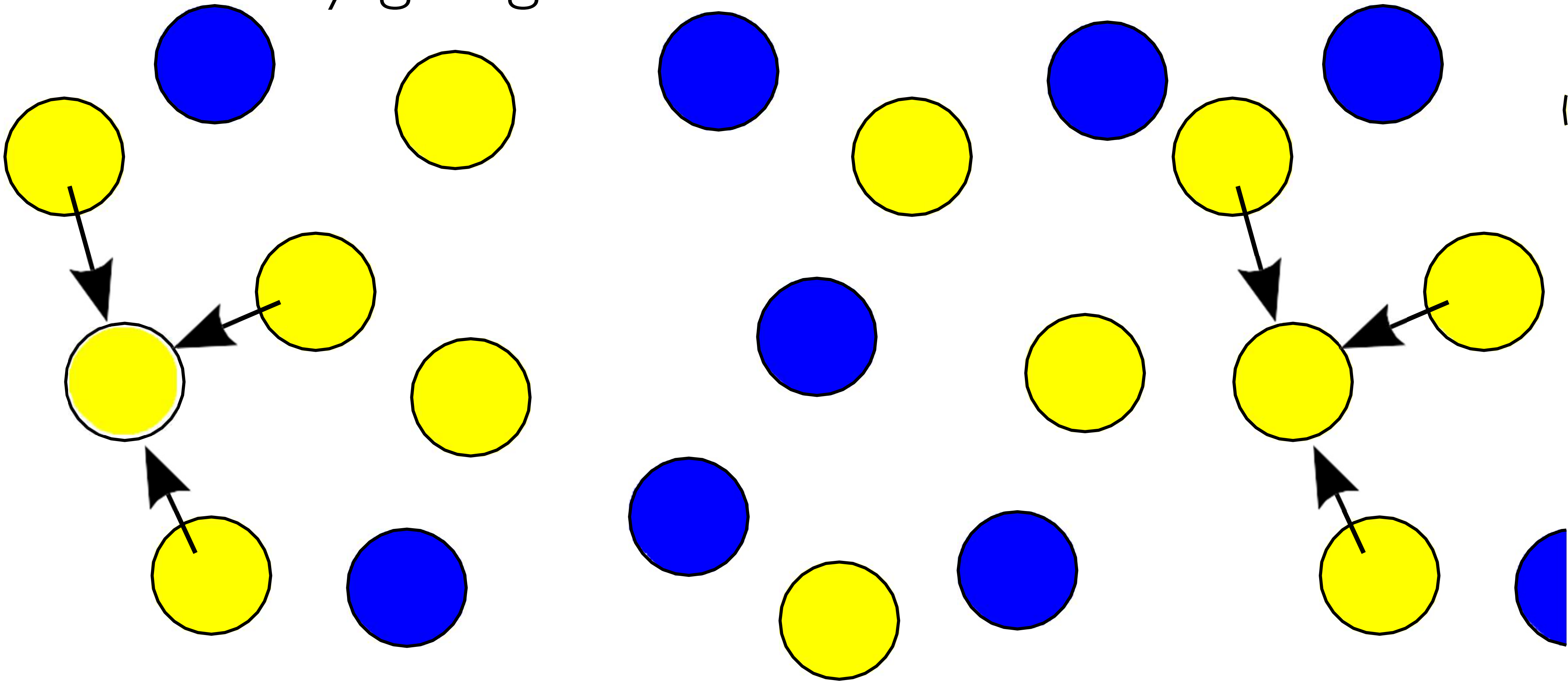


- Communication is local
- Each agent knows only about its neighbors
- No direct voting possible
- Asynchronous, local behaviors might cause deadlocks and oscillations

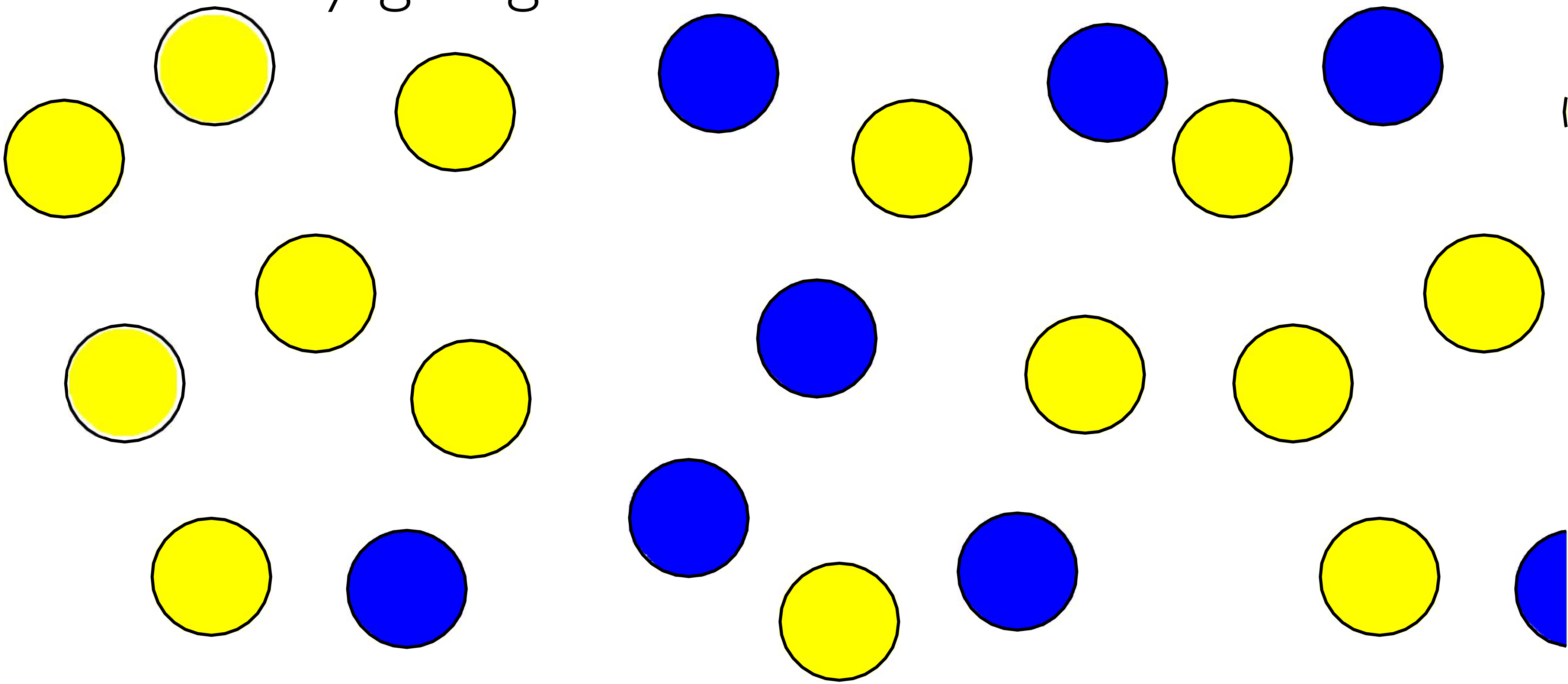
Idea: locally giving in



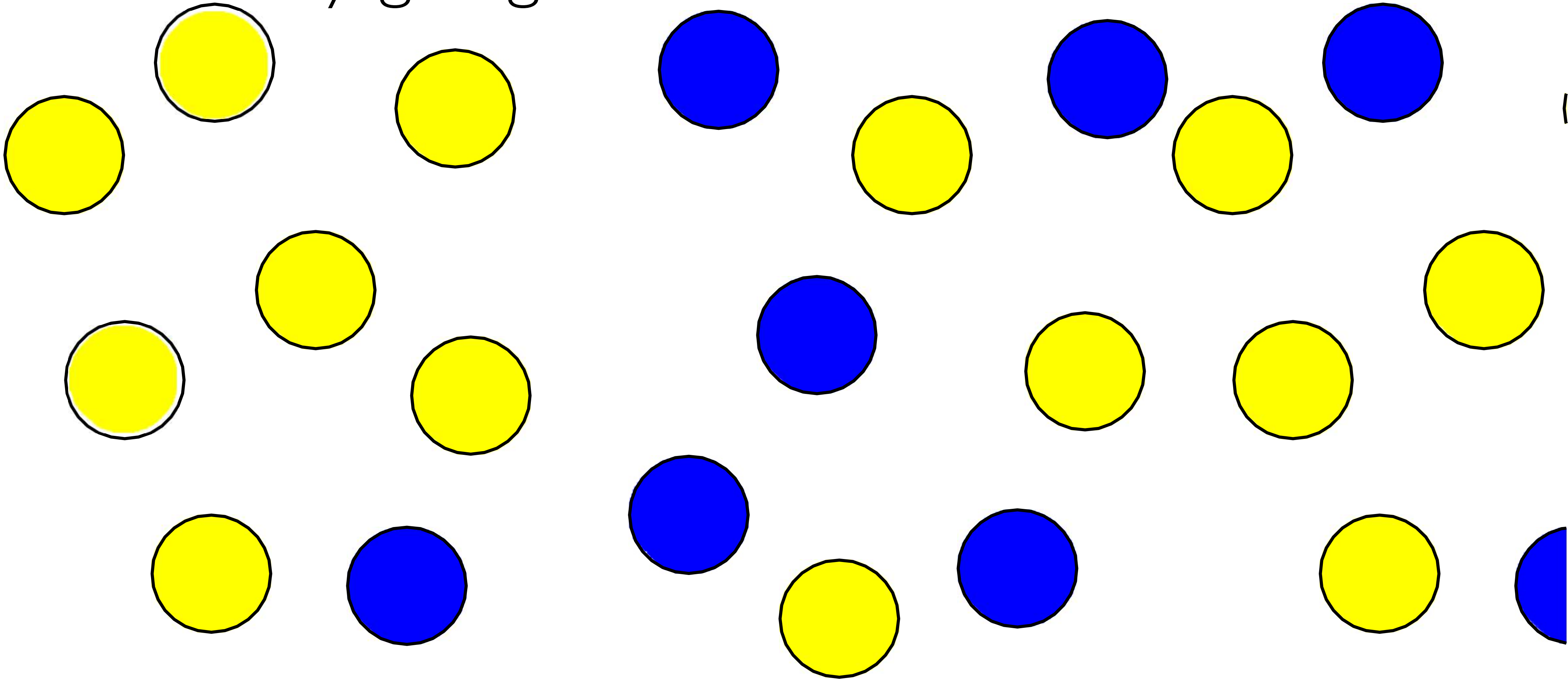
Idea: locally giving in



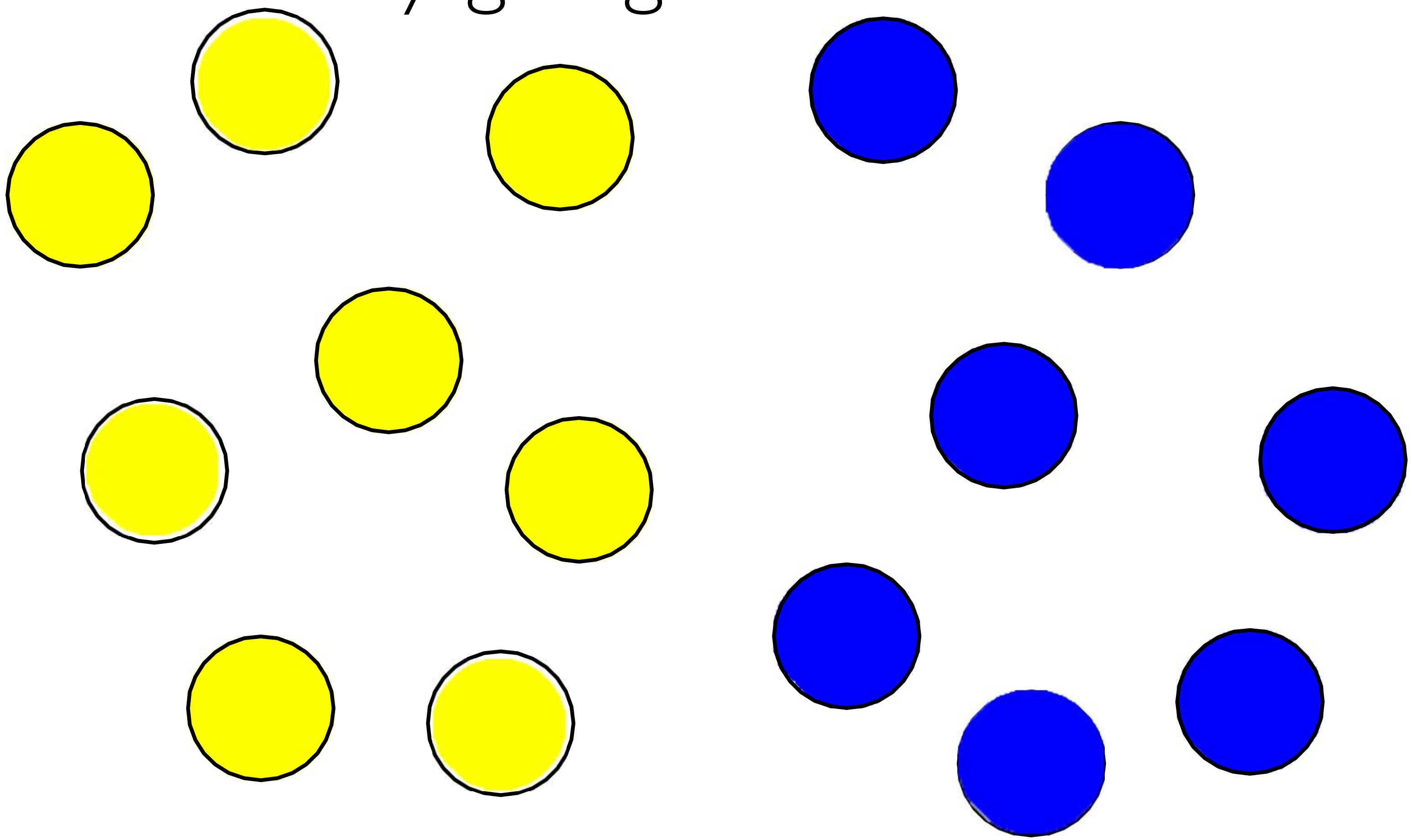
Idea: locally giving in



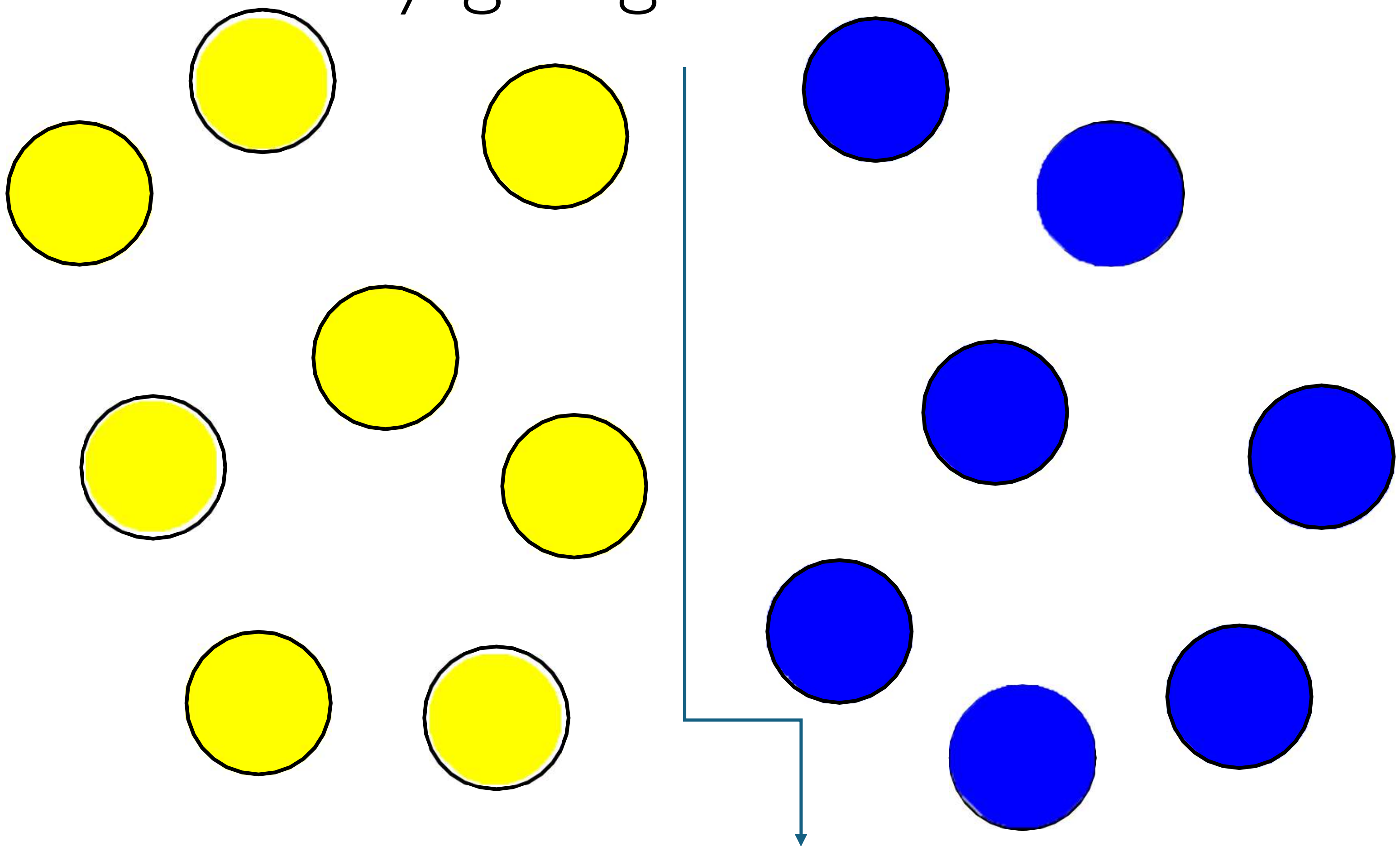
Idea: locally giving in



Idea: locally giving in



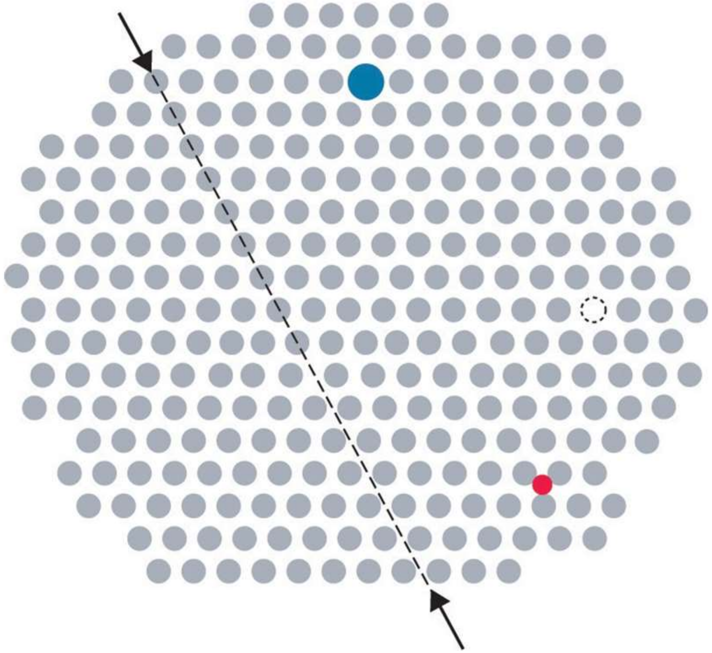
Idea: locally giving in



Comparison: Crystallization

Without knowing the global picture one part of the swarm might be doing the opposite of another part of the swarm.

an impurity like on a crystal might emerge



substitutional impurity
interstitial impurity
vacancy
line defect

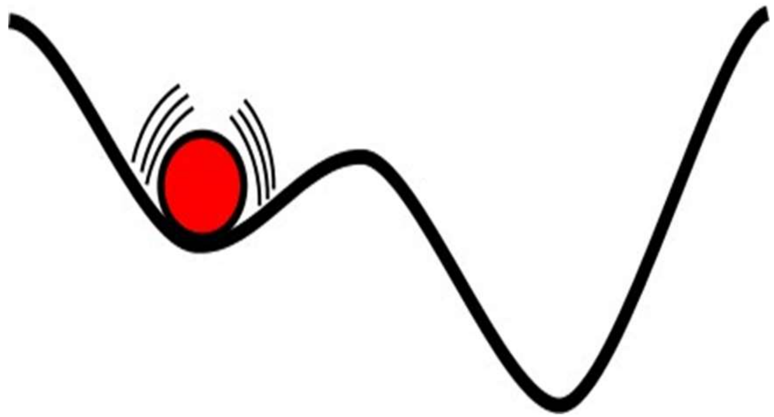
Unlike crystallization: dynamic system

Impurities will not stay forever

Because the system is dynamic and stochastic

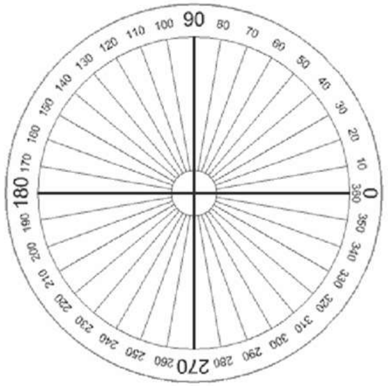
Even a tie with 50% for option A

and 50% for option B will eventually be overcome

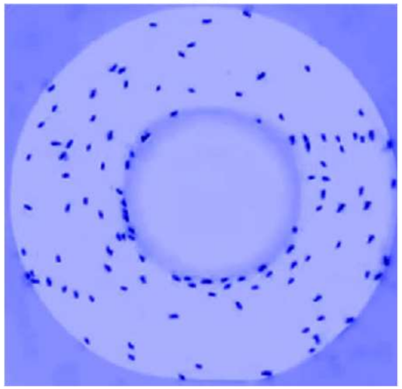


Fluctuations disturb the system and drive it out of local optima

Collective motion as decision process



A consensus on a direction can be viewed as a decision out of an infinity of alternatives [0°, 360°).

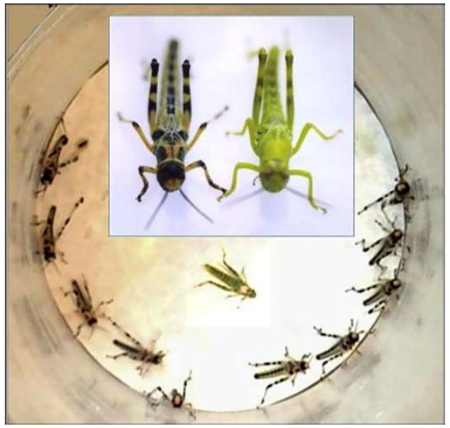


The motion can also be artificially restricted to a pseudo-1-d setting which leaves only two discrete options: clockwise motion or counter-clockwise motion.

Example: Collective motion in locusts

Observed behavior

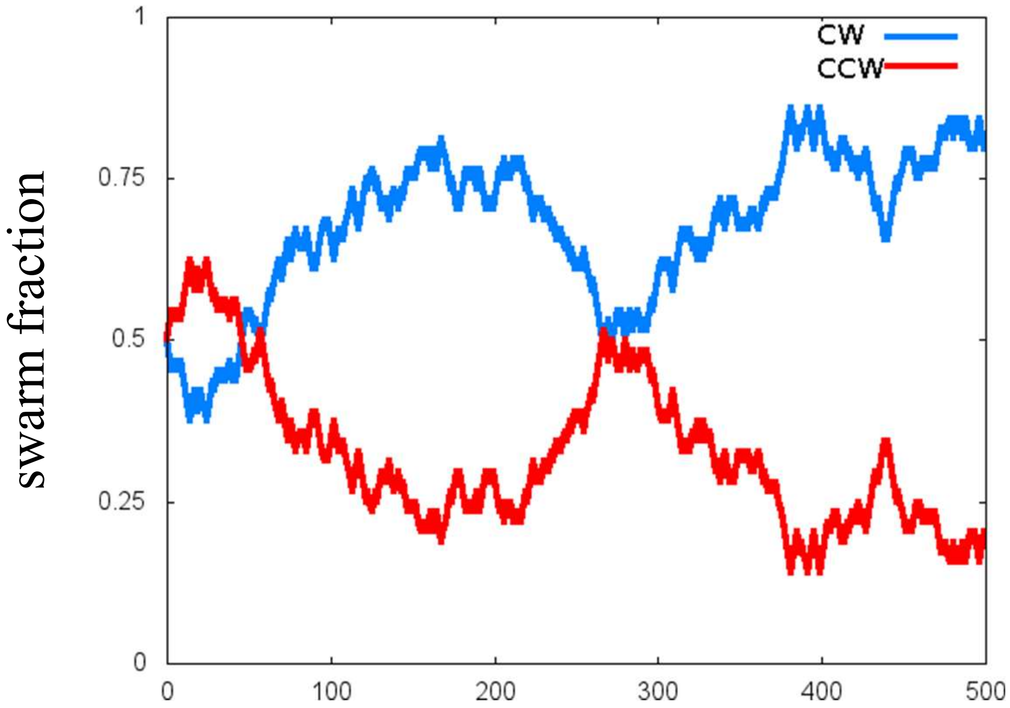
In experiments with lower locust densities groups of locust nymphs1 it was found that they were highly aligned and marched in one direction around a ring-shaped arena for up



to 2 or 3 h, before spontaneously switching direction in the space of only a few minutes and marching in the opposite direction, again for a number of hours. In experiments with higher densities, marching groups traveled in the same direction for the 8-h duration of the experiment.

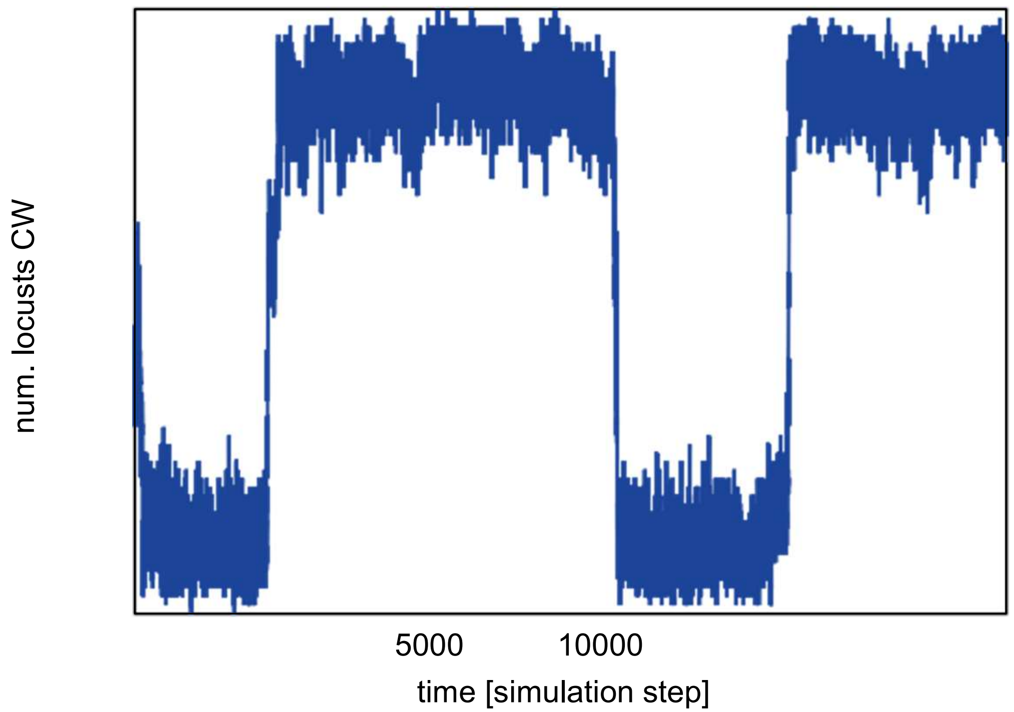
nymph: the immature form of some invertebrates, particularly insects, which undergoes gradual metamorphosis (hemimetabolism) before reaching its adult stage

Observed swarm fractions



CW: clockwise
CCW: counter-clockwise

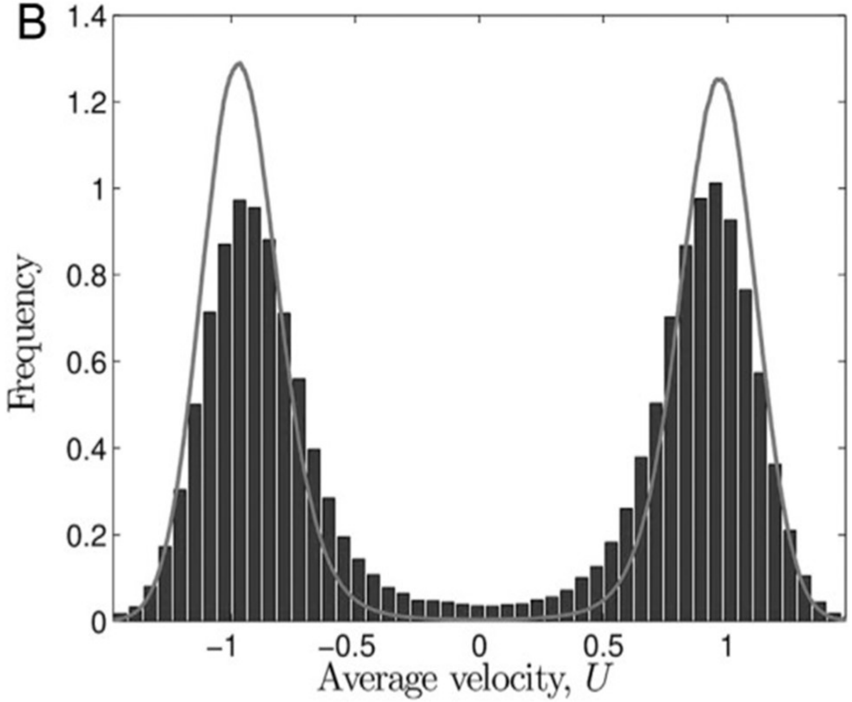
Observed switches



CW: clockwise

Distribution of velocities

The distribution of velocities measured over many independent experiments.



Inherent noise can facilitate coherence in collective swarm motion

Christian A. Yates, Radek Erban, Carlos Escudero, Iain D. Couzin, Jerome Buhle, Ioannis G. Kevrekidis, Philip K. Maini, David J. T. Sumpter
http://dx.doi.org/10.1073/pnas.0811195106

How to model this collective decision process?

- idea: keep it simple!
- binary decision process between options A and B
- say there is no bias, that is, A and B are of equal utility
- macroscopic model with only one system variable $s(t)$ which gives the swarm fraction in favor of option A (without loss of generality) at time t
- we begin with a tie: $s(0) = 0.5$ What's next?
- We are interested in how $s(t)$ changes over time depending on s itself: $\Delta s(s(t))$
- next question: How to model the process that defines Δs?

Group work

Task: model this collective decision process!

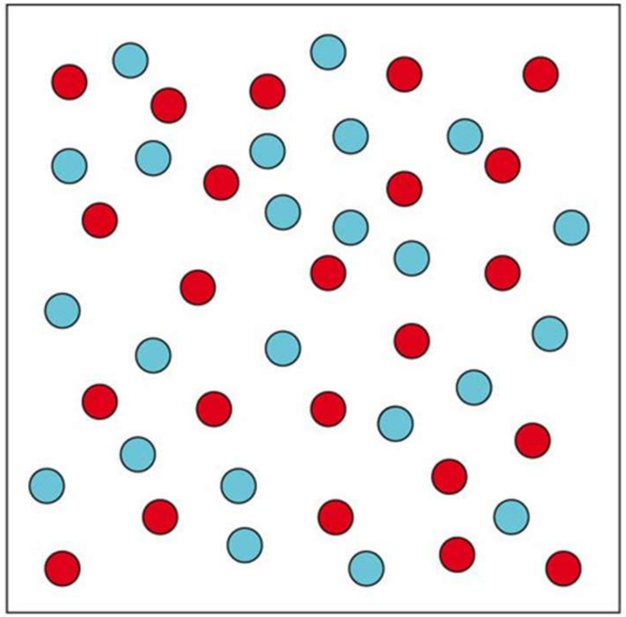
Share your ideas

Well-mixed assumption

We have a macroscopic model with just 1 state variable, the model is non-spatial (no agent positions etc.),

Hence we assume a well-mixed state.

Well-mixed means: no correlations due to spatial features



A well-mixed system. The frequency of collisions between A and B molecules is proportional to the product of N_A x N_B.

Stochastic model: Drawing

idea: well-mixed & swarm fractions ⇒ lottery drawing

robots represented by marbles in an urn (well-mixed!)

distribution of marbles according to robot states, that is, following s(t)

Ehrenfest urn model (1/2)

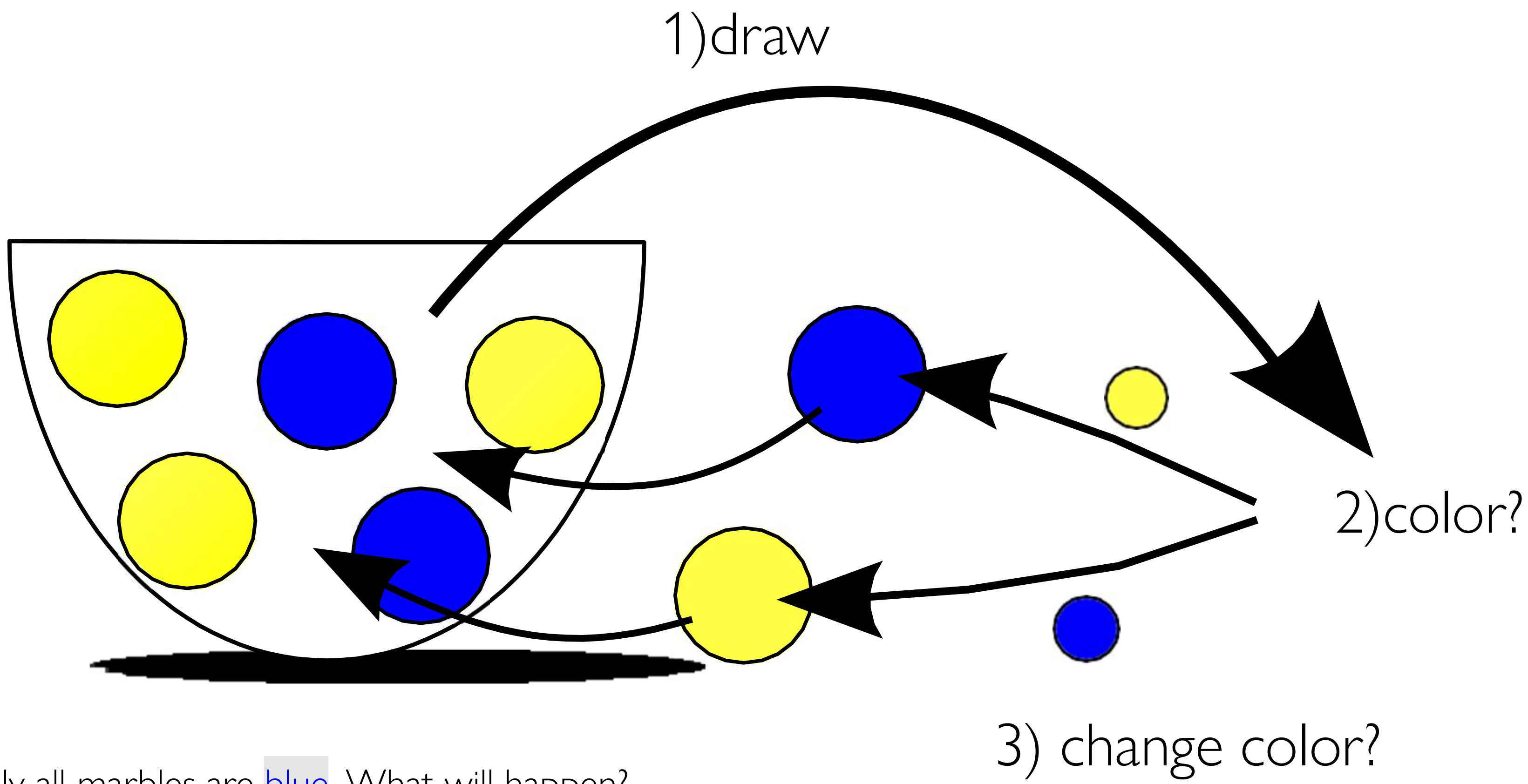
There is a famous urn model by Tatiana and Paul Ehrenfest (1907), also known as the dog-flea model.

The model is used to explain the second law of thermodynamics, in particular here: diffusion.

It works like this:

- We have an urn filled with N marbles of two colors: blue and yellow.
- Say initially all are blue. Now we start drawing. Whenever we draw a blue marble we replace it with a yellow marble. If we draw a yellow marble we replace it with a blue one.

Ehrenfest urn model: observation

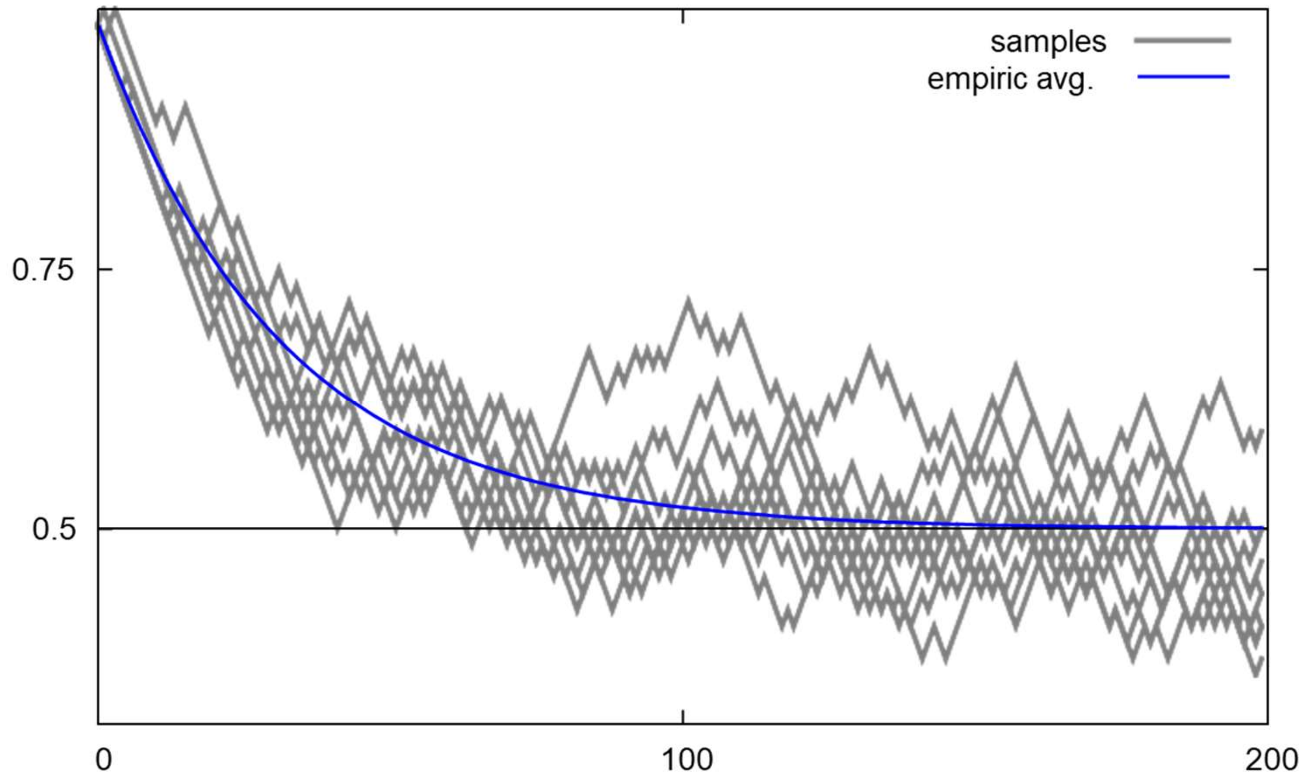


Say initially all marbles are blue. What will happen?

Ehrenfest urn model: observation

N = 64, s(0) = 1 (all blue), 200 drawings each,

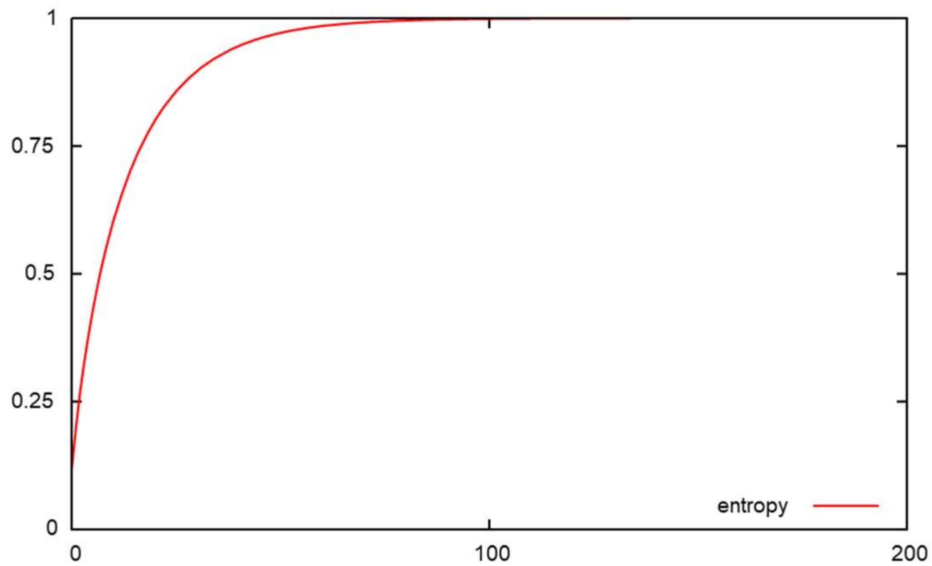
fraction of blue marbles over iterations of drawings (time):



Ehrenfest urn model: entropy

We use Shannon entropy here: $H(t) = -s(t)\log2(s(t)) - (1 - s(t))\log2(1 - s(t))$

Entropy for the empiric average over time:



monotonically increasing as proven by Mark Kac (1947)

Ehrenfest model: Relation to collective decisions?

- It is rather the opposite of a collective decision as in average it converges to the undecided state.
- Makes sense because it models diffusion while we are investigating (e.g., pattern generating) self-organizing systems.
- What to change to get a better model?
- Fluctuations should be enforced; we need to invert the model somehow. It turns out that Eigen and Winkler (1993) have done so.

Eigen urn model (1/2)

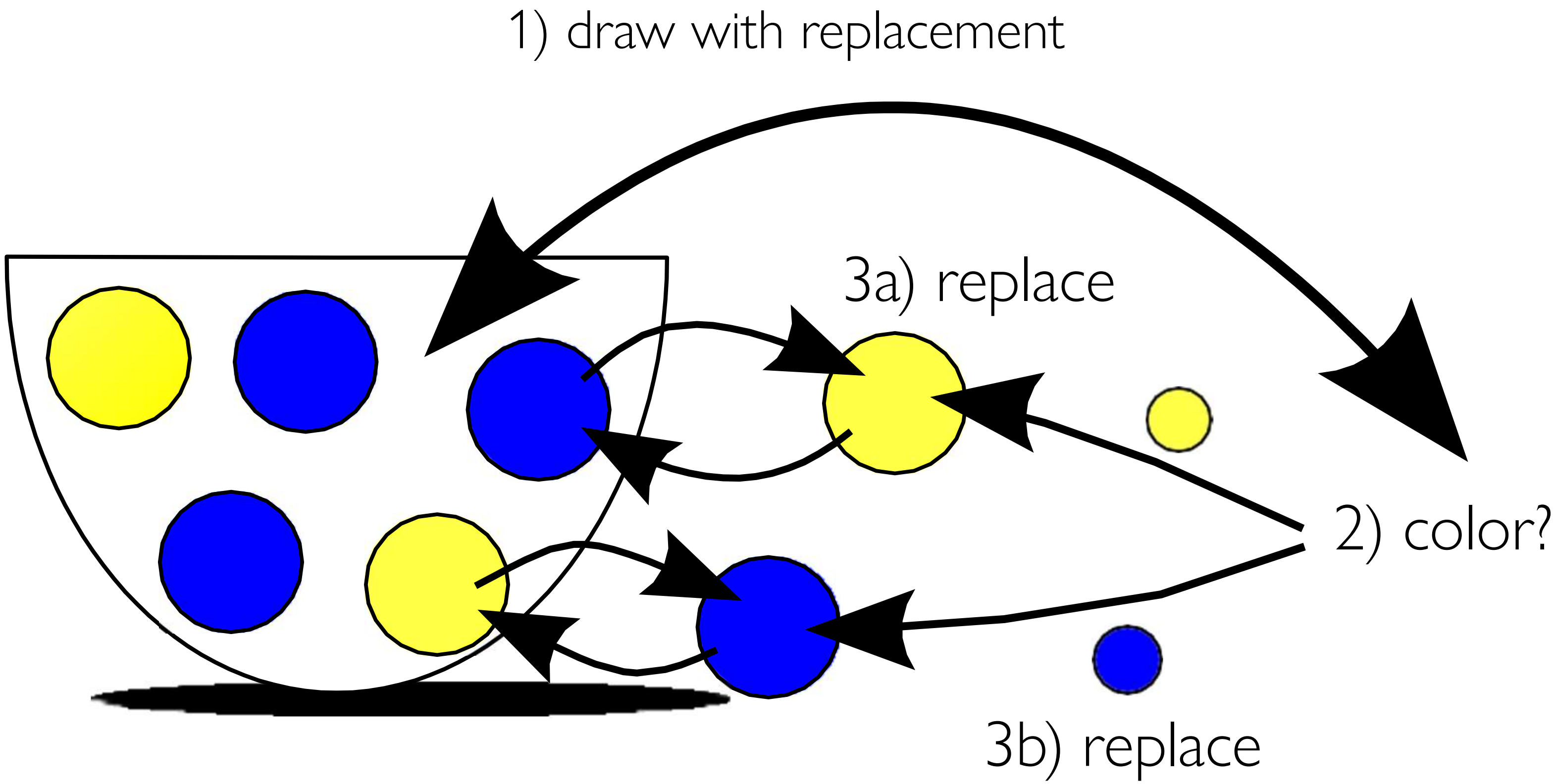
Eigen's model was defined to show the effect of positive feedback. It works like this:

We have an urn filled with N marbles of two colors:

blue and yellow.

Say initially there are 50% blue and 50% are yellow. In contrast to the Ehrenfest model we draw with replacement now. We draw a marble, notice its color, and put it back into the urn. Whenever we draw a blue marble we replace a yellow one from the urn with a blue marble. If we draw a yellow marble we replace a blue one from the urn with a yellow marble.

Eigen urn model (2/2)

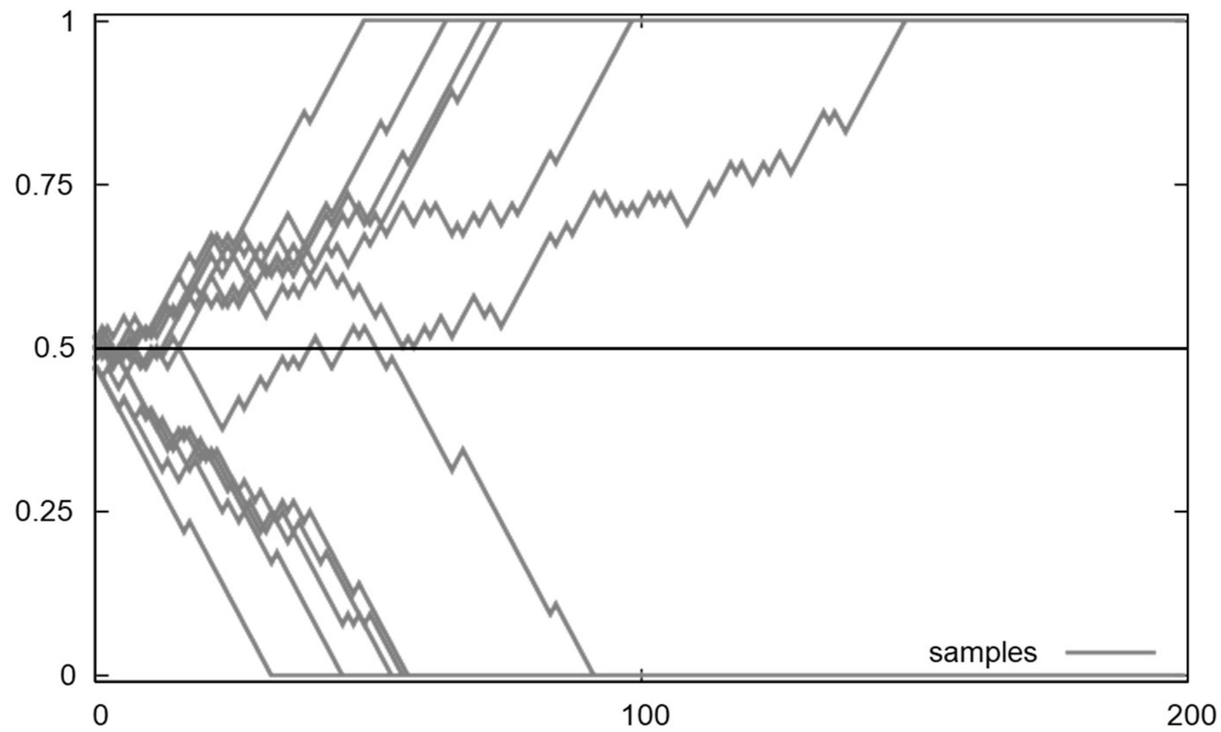


Say initially there are 50% blue and 50% are yellow. What will happen?

Eigen urn model: observation

N = 64, s(0) = 0.5 (fifty-fifty), 200 drawings each,

fraction of blue marbles over iterations of drawings (time):



Eigen urn model: Relation to collective decisions?

That looks much more like a collective decision.

What happens for $s = 0$ and $s = 1$?

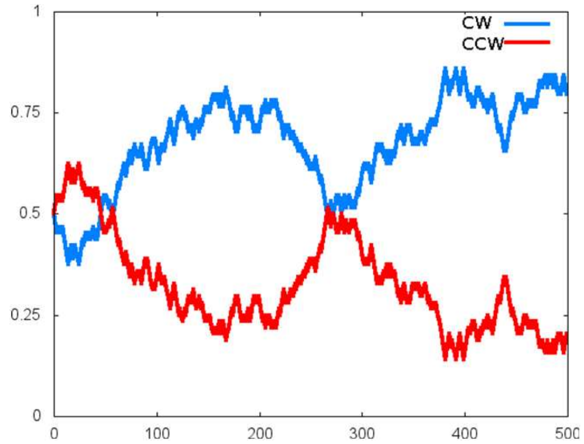
We can only draw yellow ($s = 0$) or blue ($s = 1$) marbles and would actually be asked to replace one of the other color from the urn which is impossible in these cases.

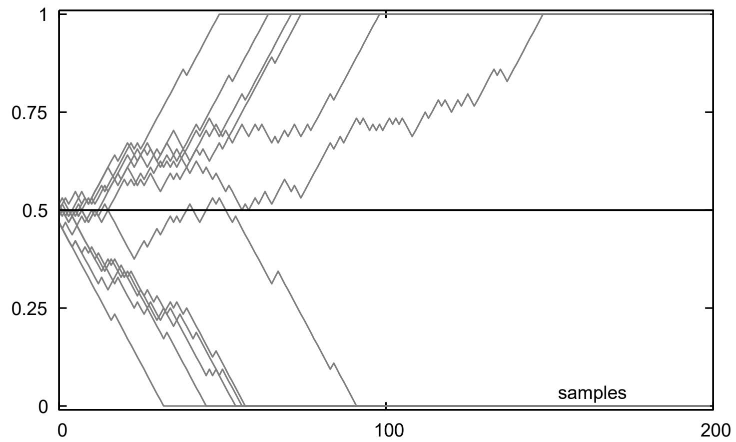
Hence, once $s = 0$ or $s = 1$ is reached we stay there forever.

Is that realistic?

Eigen urn model: Relation to collective decisions?

Locusts never reach one of the extremes (s = 0 or s = 1) and stay always dynamic





Swarm urn model: preparations

We want to come up with an extension of Eigen's model that represents collective decisions in swarms in a better way.

At first, let's think about the role of feedbacks in these systems.

Ehrenfest model: Drawing a marble of one color decreases the number of marbles of that color

⇒ negative feedback

Eigen model: Drawing a marble of one color increases the number of marbles of that color

⇒ positive feedback

What we need is a model that

has positive feedback around $s = 0.5$

and negative feedback close to $s = 0$ and $s = 1$.

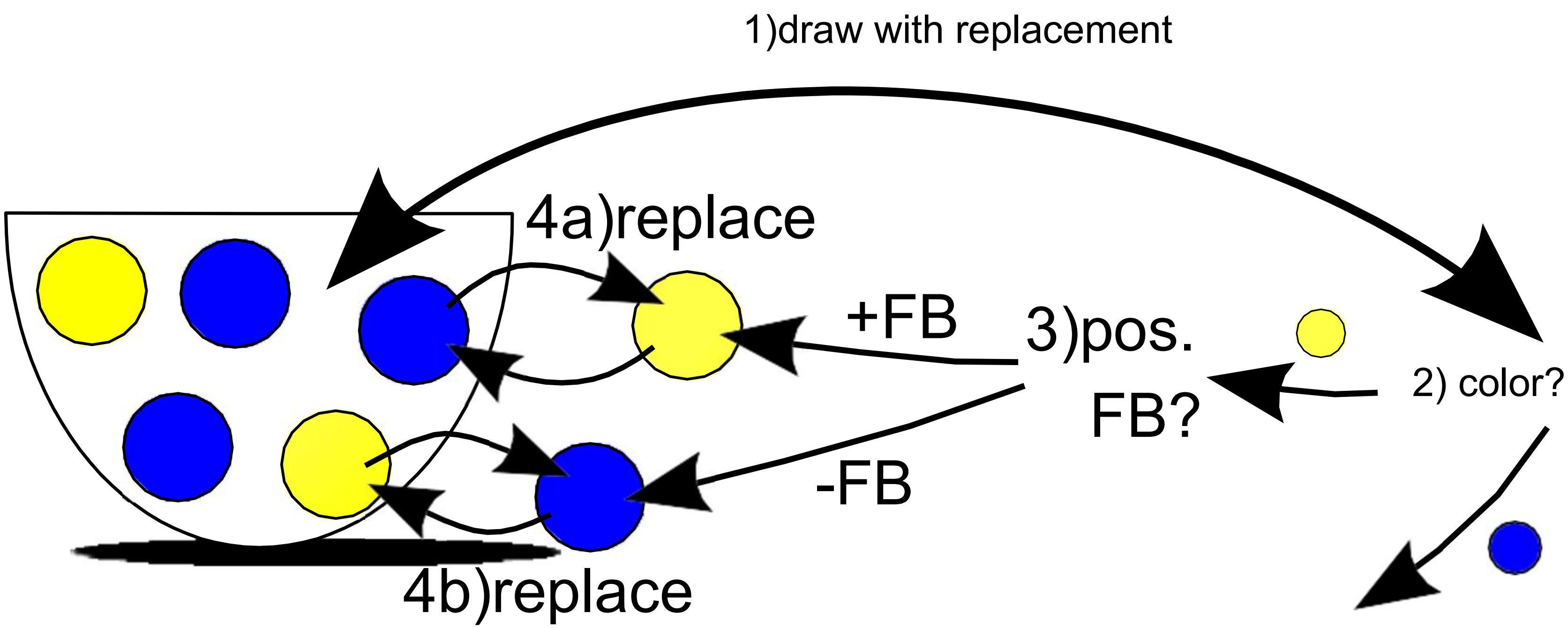
Swarm urn model

draw with replacement as in Eigen model

but explicit choice between positive and negative feedback

based on a probability for pos. feedback P_{FB}

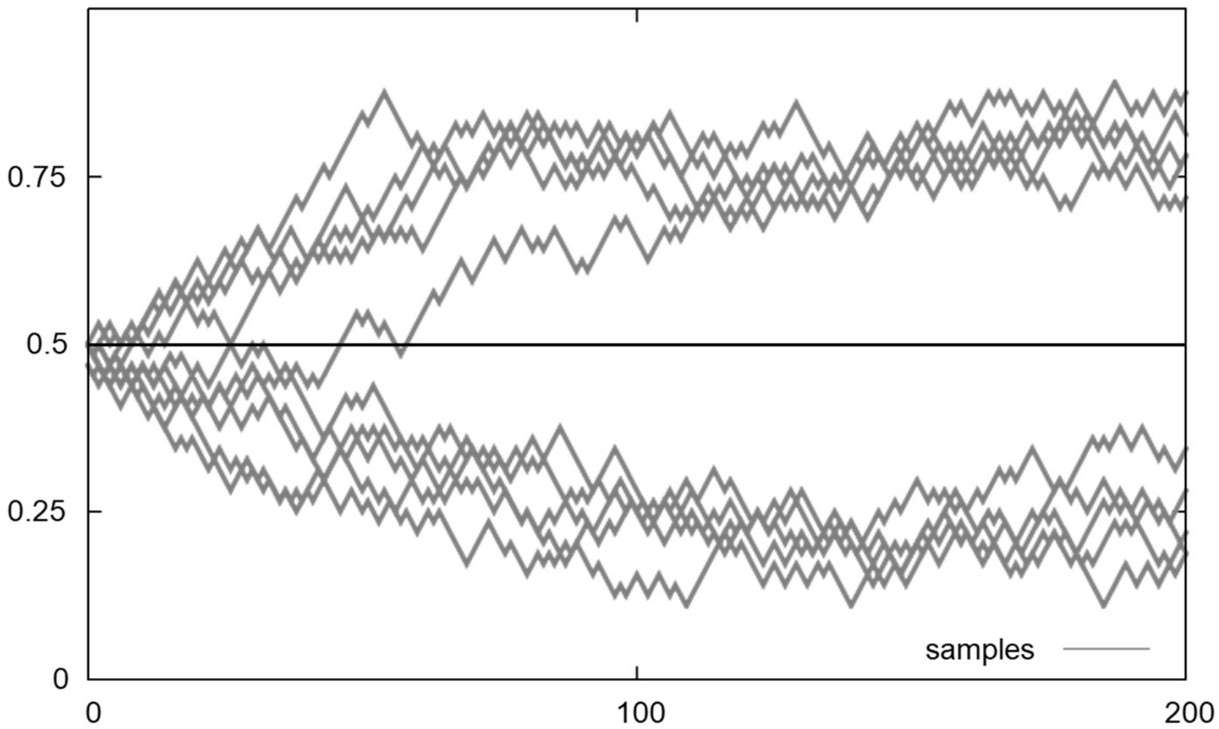
followed by the respective replacement of marbles from the urn



Swarm urn model: observation

N = 64, s(0) = 0.5 (fifty-fifty), 200 drawings each,

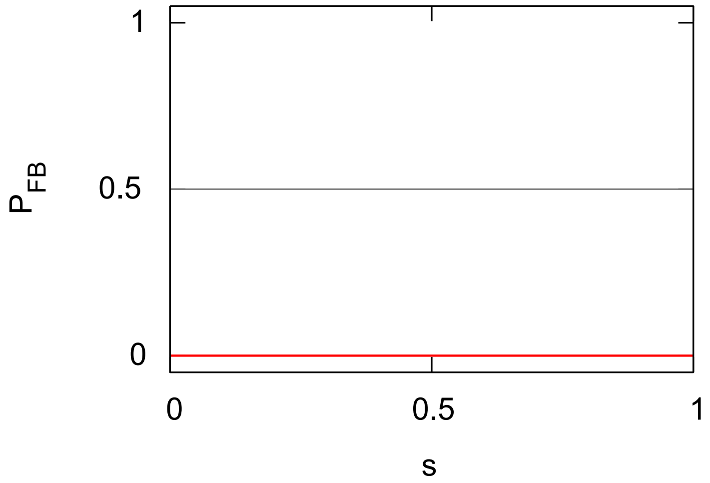
fraction of blue marbles over iterations of drawings (time):



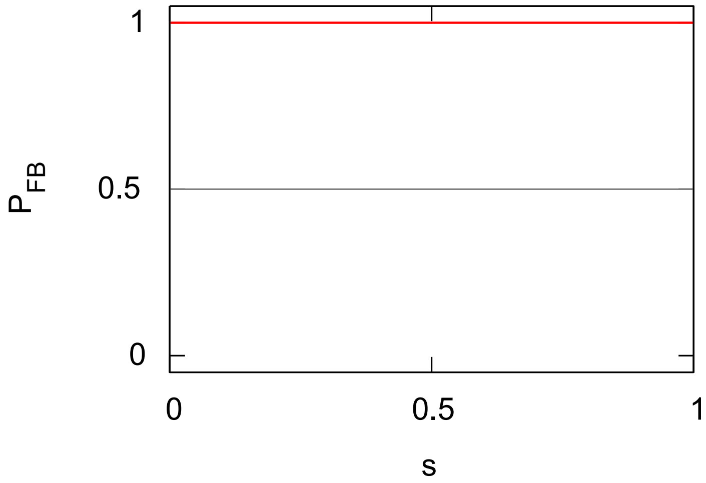
Feedbacks in Ehrenfest and Eigen model

probability of positive feedback P_{FB} in

Ehrenfest model



Eigen model

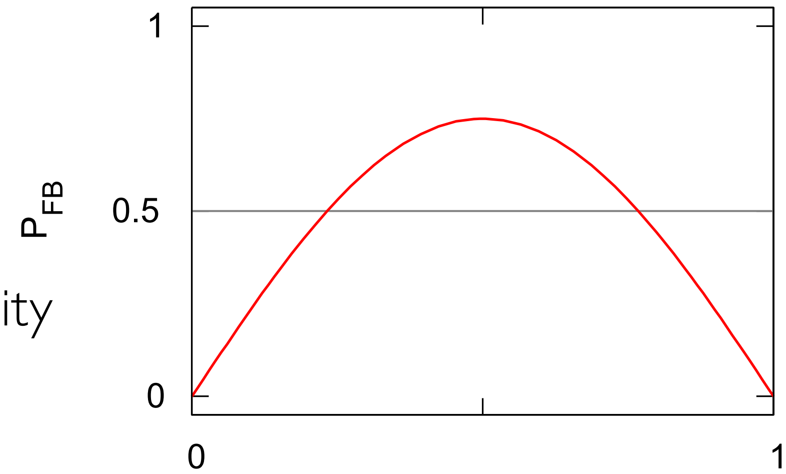


Ehrenfest model: always negative feedback

Eigen model: always positive feedback

Positive feedback probability in swarm model (1)

As said: we need positive feedback around s = 0.5 and negative feedback close to s = 0 and s = 1. an option is, for example:



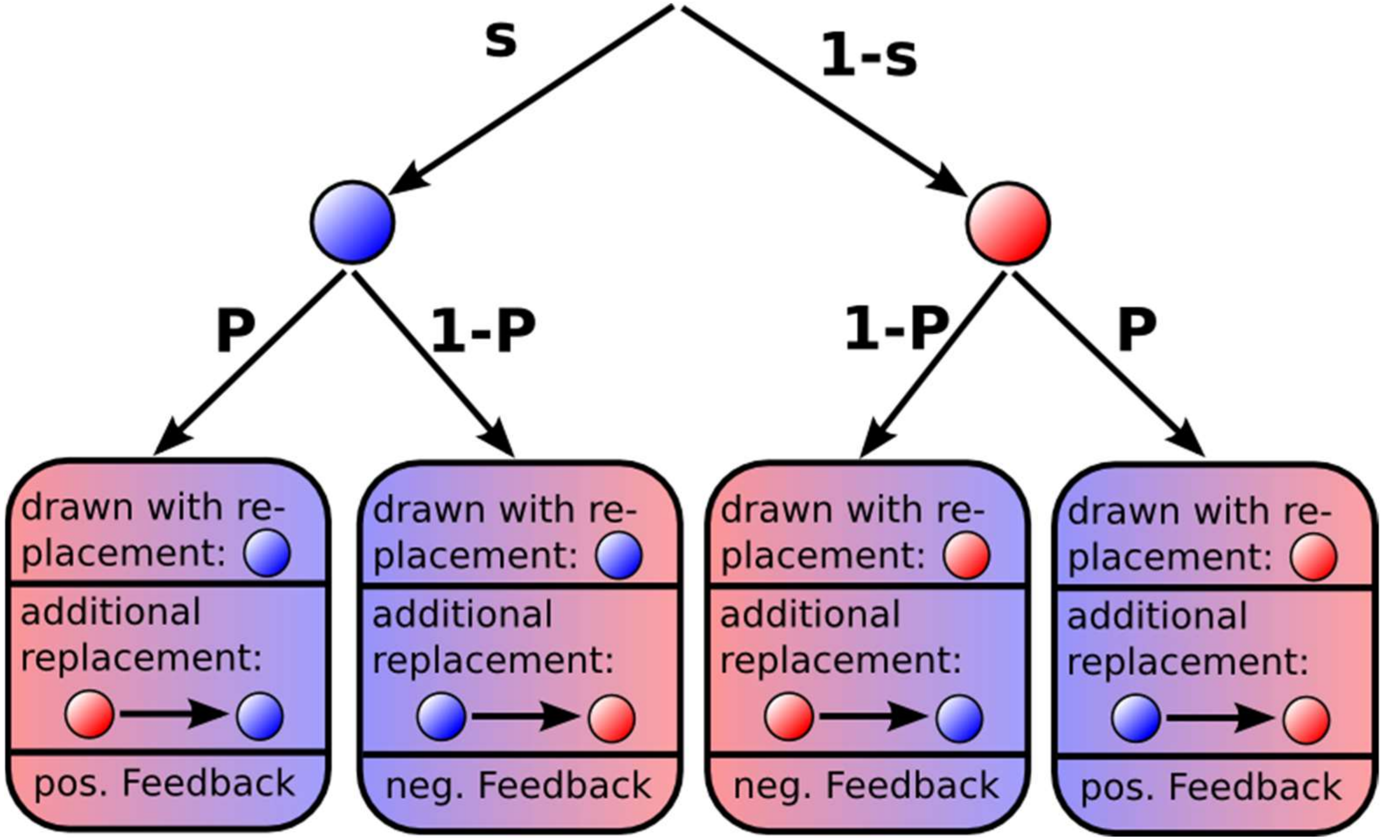
defined by $P_{FB}(s) = 0.75 \sin(\pi s)$

The general definition of the positive feedback probability

$$P_{FB}(s, \varphi) = \varphi \sin(\pi s) ,$$

for feedback intensity φ

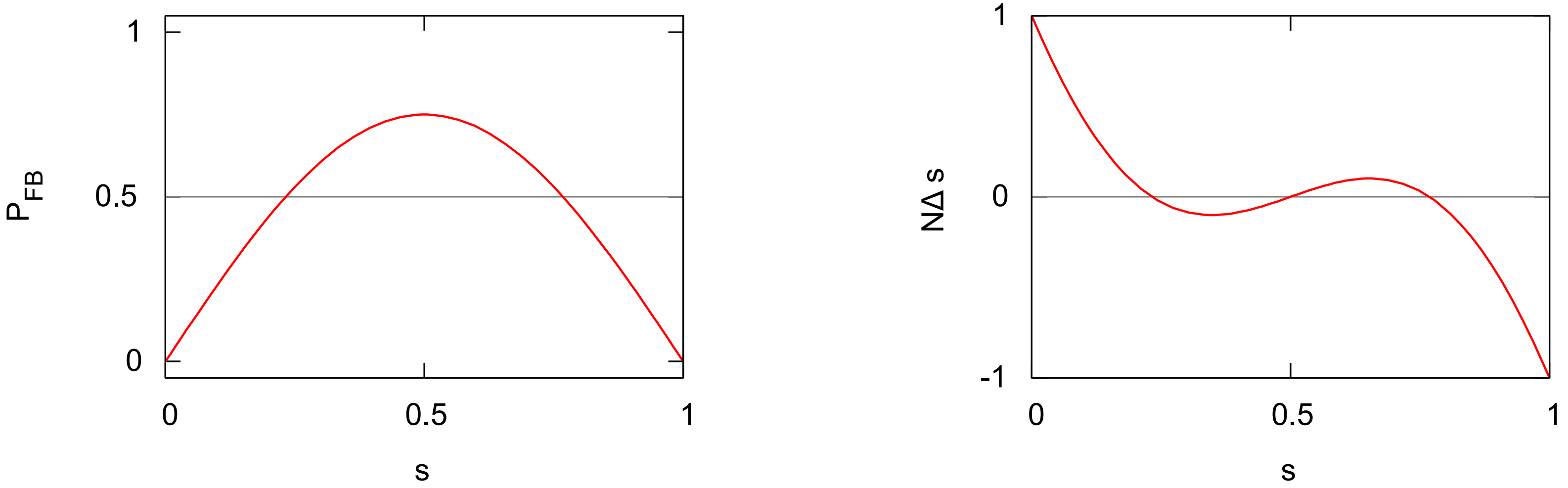
4 cases: blue/red followed by pos./neg. feedback



The complete swarm urn model

$$P_{FB}(s,\varphi) = \varphi \sin(\pi s)$$

$$\Delta s(s) = 4\left(P_{FB}(s,\varphi) - \frac{1}{2}\right)\left(s - \frac{1}{2}\right)$$



Four Cases (Probabilistic Update):

1. $s \cdot P \cdot (+1)$ → reinforce same
2. $s \cdot (1 - P) \cdot (-1)$ → switch away
3. $(1 - s) \cdot (1 - P) \cdot (+1)$ → switch to
4. $(1 - s) \cdot P \cdot (-1)$ → reinforce other

Total Expected Change:

$$\Delta s(s) = sP - s(1 - P) + (1 - s)(1 - P) - (1 - s)P$$

Expanding:

$$\Delta s(s) = 4sP - 2s - 2P + 1$$

Two minutes of thinking

1 minute to think and write one sentence

What is the highlight of lecture today for you?

1 minute to think and write one sentence

What is the challenge of lecture today for you?

References

Paul Ehrenfest and Tatiana Ehrenfest. Über zwei bekannte Einwände gegen das Boltzmannsche H-Theorem. *Physikalische Zeitschrift*, 8:311–314, 1907.

Manfred Eigen and Ruthild Winkler. *Laws of the game: how the principles of nature govern chance*. Princeton University Press, Princeton, NJ, 1993. ISBN 978-0-691-02566-7.

Stuart J. Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach*. Prentice Hall, 1995.